Chapter 8 — DVC pipelines and automation

Tool choice becomes operational when pipelines reproduce training from pinned data

Learning objectives

- Outline a live DVC workflow from init through pull for collaborators

DVC inside machine learning workflows

- Encodes transformation and training steps so experiments remain reproducible
- Teams share references rather than duplicating large files

Example 8.25 — DVC in an ML Pipeline

- Example 8.25 — hands-on module
- Storage burden stays on remotes
- Explore the chapter example module
- View files: `modules/chapter8/example25/`

Pipeline stages as executable history

- A DVC pipeline declares stages with commands, dependencies
- When inputs or code change
- That dependency graph is the operational heart of reproducibility

Example 8.28 — DVC Pipeline Stages for Training

- Example 8.28 — hands-on module
- Example 8.28 defines preprocessing, training
- Explore the chapter example module
- View files: `modules/chapter8/example28/`

Example 8.28 — listing

```
stages:
  preprocessing:
    cmd: python preprocess.py data/raw_data.csv
data/preprocessed_data.csv
    deps:
      - data/raw_data.csv
    outs:
      - data/preprocessed_data.csv
  training:
    cmd: python train_model.py data/preprocessed_data.csv
models/model.pkl
    deps:
      - data/preprocessed_data.csv
    outs:
      - models/model.pkl
  evaluating:
    cmd: python evaluate_model.py models/model.pkl
```

Continuous integration for dataset updates

- Trigger training when code or data lands on a protected branch
- Automation reduces forgotten manual steps and keeps shared remotes current for the team’s

Example 8.29 — Live Demonstration of DVC Automation

- Example 8.29 — hands-on module
- Example 8.29 outlines a live walkthrough
- Explore the chapter example module
- View files: `modules/chapter8/example29/`

Example 8.29 — listing

```
git init
dvc init
dvc remote add -d myremote s3://mybucket/datasets
```

Takeaways

- DVC pipelines turn versioned data into reproducible workflows
- Stage YAML records commands, dependencies, and outputs
- Together they close the gap between “we have a dataset” and “we can retrain from the same

Next

- Complete the quiz for this part
- The next part turns documentation and versioning into day-to-day operating practice